



UNIVERSITÀ
DI TRENTO

Dynamic content: the server side programming

1

Till now...

Web servers like to serve static pages:

- They just stay on a folder in the server
- They are just sent to clients
- Every client sees the same thing

Sometimes we want more....

A simple problem

what is the current time?



3

Dynamic pages

e.g. what is the current time?

We want to obtain *NON STATIC* information from the server

```
<html>
<body>
The current time is always
1 PM on the server
</body>
</html>
```

```
<html>
<body>
The current time is [InsertTime]
on the server
</body>
</html>
```

This implies executing some code on it, and send the results to the client.

We name these page dynamic (or just-in-time) pages: they are dynamically-created pages not existing before the request

4

A first approach

- The web server run a process that compute what you need and generate (write!) a HTML page
- The page is inserted in the response body and sent to the client
- It is called Common Gateway Interface (CGI)

```

1 #echo off
2
3 echo Content-type: text/html
4 echo.
5
6 set mydate=%date%
7 set mytime=%time%
8
9 echo <!DOCTYPE html>
10 echo <html lang="en">
11 echo   <head>
12 echo     <meta charset="utf-8">
13 echo     <meta name="viewport" content="width=device-width, initial-scale=1.0">
14 echo     <title>What time is it?</title>
15 echo   </head>
16 echo   <body>
17 echo     <p>Hello, I am a perl script</p>
18 echo     <p>Today is %date%</p>
19 echo     <p>and time is %time%</p>
20 echo   </body>
21 echo </html>

```

Perl

```

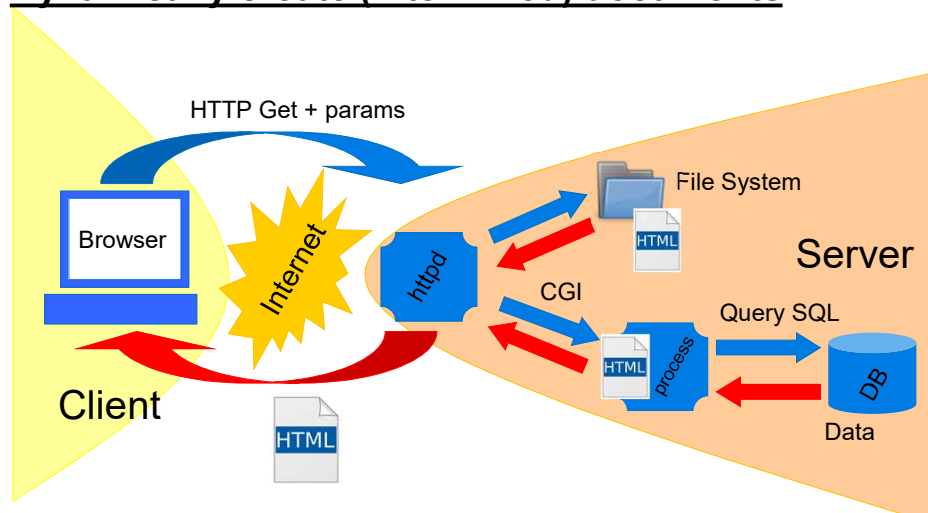
#!C:\xampp\perl\bin\perl.exe
print "Content-type: text/html; charset=iso-8859-1\n\n";
($sec,$min,$hour,$mday,$mon,$year,$yday,$dow) = localtime();
print "<!DOCTYPE html>";
print "<html lang='en'\>";
print "<head>";
print "<title>What time is it?</title>";
print "</head>";
print "<body>";
print "<p>Hello, I am a perl script </p>";
print "<p>Today is %date%</p>";
print "<p>and time is %time%</p>";
print "</body>";
print "</html>";

```

Bat

5

Evolution 1: Dynamically create (interlinked) documents



Introducing “Web server languages”

7

A simpler idea

Instead of embedding HTML into the code, we could embed the code into HTML...

Template page

```
<HTML>
SOME HTML STUFF
%SOME CODE%
MORE HTML STUFF
</HTML>
```

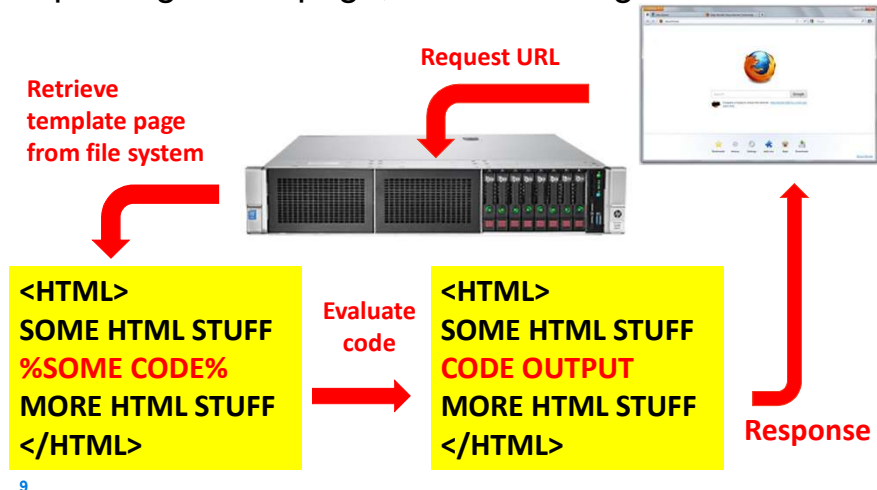


```
<HTML>
SOME HTML STUFF
CODE OUTPUT
MORE HTML STUFF
</HTML>
```

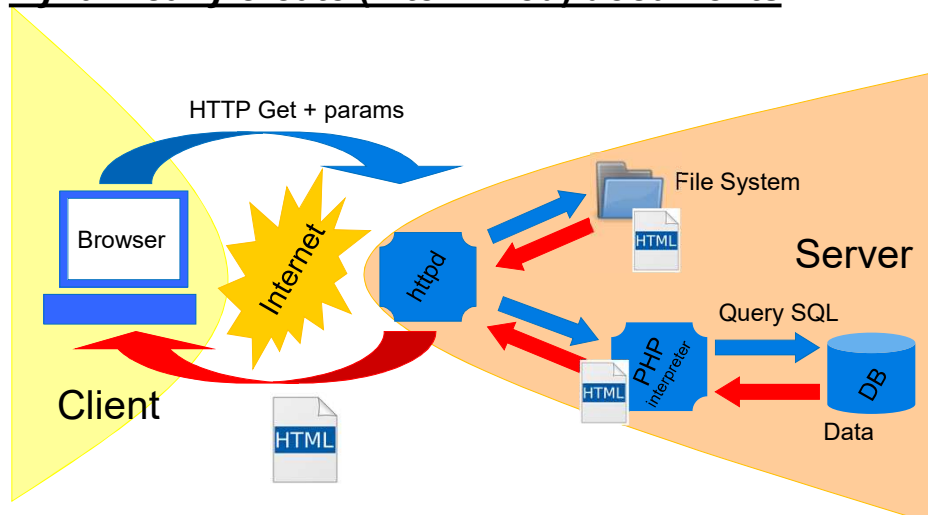
8

A simpler idea

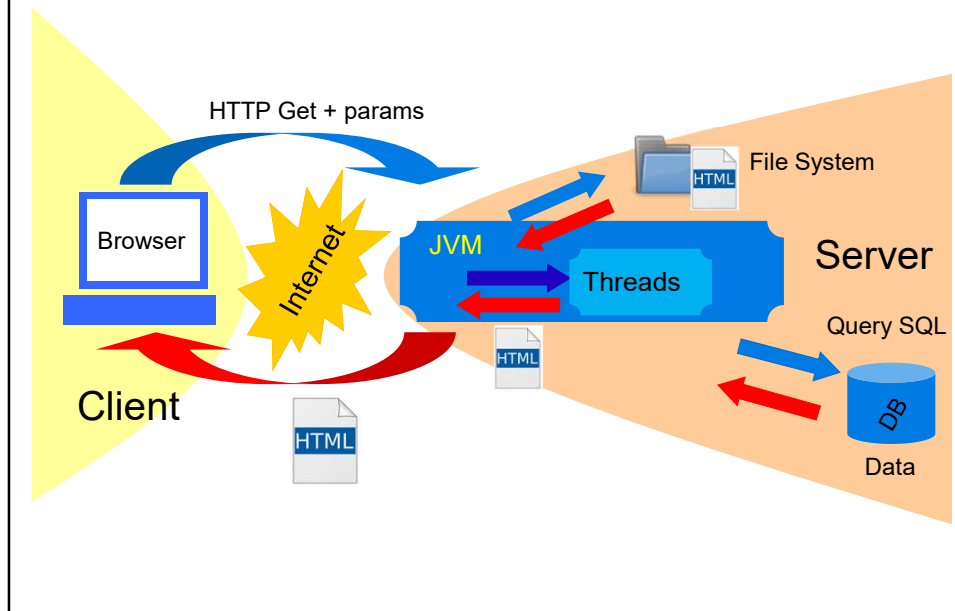
Augment the web server with an engine capable of parsing a web page, and executing code in it



Evolution 1: Dynamically create (interlinked) documents



Evolution 1: Dynamically create (interlinked) documents

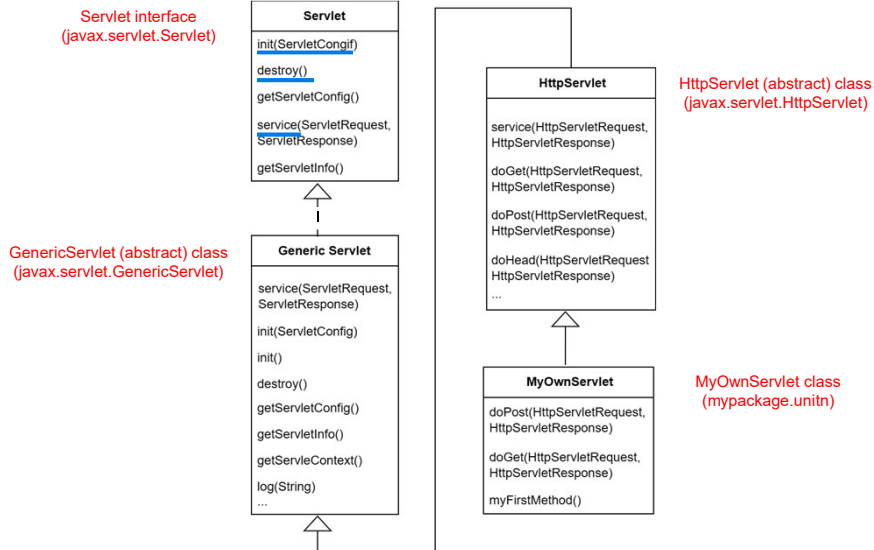


A closer look

- A Container is used to understand HTTP and translate the request / response HTTP to a Java object, so that it does not need to take care of implementing the communication layer
- Such a kind of Container is called a Servlet Container

A servlet is a Java class that directly interacts with the Container, handling requests (not necessarily http requests), translating them in Java objects, processing them, and replying back with a response

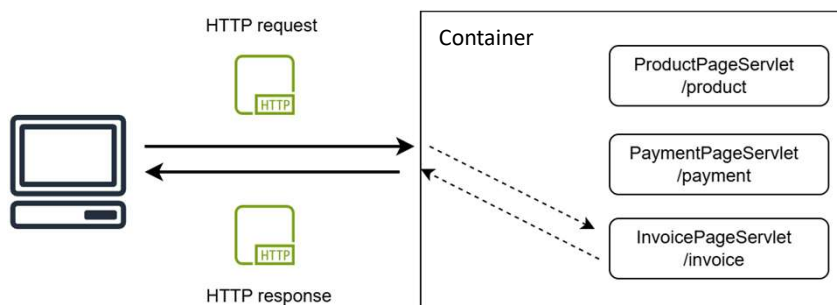
A pseudo-UML of Servlet



13

Something more yet

- Servlets are the most critical components of backend

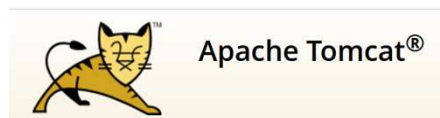


- Each servlet is associated with a path. When a client sends a requests, the Container calls the servlet registered at that specific path. The servlet gets the values on the request and builds the response that the Container sends back to the client

14

Apache Tomcat

- Apache Tomcat is a Java servlet container, and is run on a Java Virtual Machine (JVM).
- Tomcat utilizes the Java servlet specification to execute servlets generated by requests, often with the help of JSP pages, allowing dynamic content to be generated much more efficiently than with a CGI script.



<https://tomcat.apache.org/>

15

HttpServletRequest and HttpServletResponse

- As you know, when the Container receives requests from a client, it creates two Java objects:
 - **HttpServletRequest**
 - **HttpServletResponse**

and passes them to the invoked servlet

16

A pseudo-UML of HttpRequest and HttpResponse

